Recap

a relation from A to B is a subset of $A \times B$

When the relation is named R,

we can write aRb to mean $(a, b) \in R$

In particular a function

$f : A \to B$ can be identified with the set

$\{(x, y) \mid f(x) = y\}$

and thus a function is a relation.

Since a relation is just a set, we can apply set operations

$\cap, \cup, \setminus, \complement$

are all relevant.

Example

on $\mathbb{R}$ we can define

$R_1 = \{(x, y) \mid x \leq y\}$

$R_2 = \{(x, y) \mid x < y\}$

$R_3 = \{(x, y) \mid x \geq y\}$

$R_4 = \{(x, y) \mid x = y\}$

$R_1 \cap R_3 = R_4$

If $(x, y) \in R_1 \cap R_3$

then $(x, y) \in R_1$ so $x \leq y$

but also $(x, y) \in R_3$ so $x \geq y$

but if $x \leq y \land y \leq x$ then $x = y$

so xR_4y

that shows $R_1 \cap R_3 \subseteq R_4$

Also if

xR_4y then $x = y$

so $x \leq y$ and $y \leq x$

thus $(x, y) \in R_1$ and $(x, y) \in R_3$

hence $(x, y) \in R_1 \cap R_3$

so $R_4 \subseteq R_1 \cap R_3$

Similarly

$\overline{R_2} = R_3$

If $(x, y) \in \overline{R_2}$ then $(x, y) \notin R_2$

so $\neg(x < y)$ thus $x \geq y$ and xR_3y

similarly if

$(x, y) \in R_3$ then $x \geq y$ so $\neg(x < y)$ and $x\not{R_2}y$

Exercise

Check $R_1 \setminus R_2 = R_4$

what is $R_1 \cup R_3$?

We can also think of relations like functions and build composition

Example

Let S be the set of students

V be the set of video games

C be the set of characters

Two relations, P and F might encode data

P from S to V is defined by xPy if student x has played game y

F from V to C is defined by yFz if game y features character z

If a marketing team wants to determine which students have played a game featuring a particular character, the relations are enough information.

We would say "x has played a game featuring character z" (Some relation between S to C)

If $\exists y \in V$ such that,

xPy and yFz

The composite relation is $F \circ P$ has this definition.

More generally $aT \circ Ub$

equivalently $(a, b) \in T \circ U$

iff $\exists c$ such that aUc and cTb

There is an analogy to functions,

$g \circ f(x) = z$

means $g(f(x)) = z$

so if we name $f(x)$ as y, then

$\exists y f(x) = y$ and $g(y) = z$

If R is a relation on A

(a relation from A to A)

then we can compose R with itself

In fact we can repeat and build a recursive definition

$R^1 = R$ and $R^{n+!} = R^n * R$

Example

Suppose S is the relation on $\mathbb{N}$ given by xSy iff $x < y$

$4S^26$

since $4S5$ and $4S6$

but $(8, 9) \notin S^2$ since there are no numbers $m\mathbb{N}$ strictly between 8 and 9

Example 2

let A be the set of bus stops in Ontario, and say xRy iff we would take the bus from x to y

If xR^2z then

$\exists y$ such that xRy and yR^2z

equivalently xR^2z iff we can get from x to z by taking exactly 2 busses. (With transfer at y for some y)

xR^kz iff you can get from x to z using exactly k busses.

To study the trips possible by any sequence of busses, we would consider

$R \cup R^2 \cup R^3 \cup ... \cup R^k \cup ...$

Sometimes denoted R^* (The transitive closure)

or

$\bigcup_{k=1}^{\infty} R^k$

Some properties

(of relations on a set)

(in particular that make a relation on “like” the “=” relation)

Motivating examples

For real numbers,

of $x \leq y$ and $y \leq z$ and $z \leq x$

then

$x \leq z$

but this means $x \leq z$ and $z \leq x$

so $x = z$

For sets

If $S \subseteq T$ and $T \subseteq U$ and $U \subseteq S$

then $S \subseteq U$ and $U \subseteq S$

thus $S = U$

For propositions

If $p \to q$ and $q \to r$

and $r \to p$, then

$p \to r$ (Hypothetical Syllogism)

and thus $p \to r$ and $r \to p$

so $p \leftrightarrow r$ and

$p \equiv r$

Our properties describe why these arguements are so similar.

All of $\leq, \subseteq,$ and $\to$

are transitive

while $\leq,$ and $\subseteq$ are antisymmetric

$\to$ is not antisymmetric, hence the conclusion

$p \equiv r$ instead of $p = r$

Definition

For a relation R on set A

1. R is reflexive iff $\forall x \in A(xRx)$ ($=, \leq, \subseteq, \to$ are reflexive, $<$ is not)
2. R is symmetric iff $\forall x \in A \forall y \in A(xRy \to yRx)$ ($=$ is symmetric, $\leq$ is not)
3. R is antisymmetric iff $\forall x \in A \forall y \in A(xRy \wedge yRx \to x = y)$
4. R is transitive iff $\forall x \in A \forall y \in X \forall z \in A(xRy \wedge yRz) \to xRz$ ($\leq, <, =, \to$ are transitive)

Aside

Hypothetical Syllogism is the rule that states $\to$ is transitive.

In our bus example R is not transitive, there are places I can get with transfer that I cant reach directly.

Example

On the set $\{2, 4, 7, 14, 28\}$

define xRy by xy is a factor of 28

$2 \to 2$

$14 \to 2$

$7 \to 4, 2$

This is symmetric, every arrow has a corresponding opposite direction.

It is not reflexive, since there is no loop at 4,

Example

on the set $\{2, 3, 4, 5, 6, 7, 8\}$

xRy iff x is a factor of y

$2 \to 2, 4$

$3 \to 3, 6$

$4 \to 4, 8$

$5 \to 5$

$6 \to 6$

$7 \to 7$

$8 \to 8$

This is reflexive, but not symmetric, it is antisymmetric and transitive